

Maanas Tyagi

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PROFESSIONAL SUMMARY

AI engineer and practitioner building production automation systems for Fortune 500 clients at Genpact. Deep expertise in Retrieval-Augmented Generation (RAG), and agentic AI architectures. Two on large language model (LLM) behavior — one published, one under review.

PROFESSIONAL EXPERIENCE

AI & Analytics Intern

May 2026 – Present

Genpact · Jaipur, India

- Replaced a \$3,000 licensed reconciliation software with a custom Python automation tool handling files up to 10,000 rows — reduced processing time from 30+ minutes to under 5 minutes
- Built GRNI vs PO reconciliation engine for enterprise clients Otis and Polestar — eliminated manual pivot table creation for Chartered Accountants
- Developing purchase reconciliation software; Accountant upload a single file and receive fully cross-matched output instantly

PUBLICATIONS

- Instruction Fidelity Decay in Small Language Model Agents: Semantic Recall Dissociates from Format Compliance Under Semantic Interference — under review
- Investigated how agents degrade when a Small Language Model (SLM) acts as the reasoning brain under varying levels of semantic noise and context interference
- Front-Loading Bias in Large Language Models: A Multi-Model Empirical Analysis — ResearchGate preprint (DOI: 10.13140/RG.2.2.32504.92164)
- Originated from a personal observation: in multi-point LLM replies, models like ChatGPT consistently generate fewer words per point as the response grows — formalized into an empirical study across GPT, Gemini, and Claude

Both papers grew from independent observations and curiosity, developed without formal research supervision.

PROJECTS

Lumen Research *FastAPI · React · FAISS · Ollama · RAG · Python*

- Local-first AI research paper reader — upload PDFs up to 100+ pages, chat with content, highlight text for ELI5 / mathematical / step-by-step explanations
- RAG pipeline: PyMuPDF extraction → sliding window chunking → Ollama embeddings → FAISS cosine similarity → streamed LLM response
- Zero cloud dependency via Ollama; deep-dive threads with isolated memory, auto-summaries, and multi-mode explanations

Corrective RAG (CRAG) *LangGraph · LangChain · Python · FAISS*

- Implemented full Corrective Retrieval-Augmented Generation (CRAG) pipeline with document grading, web fallback, and query rewriting
- State machine: retrieval → grade → correct/rewrite → generate; handles retrieval failure gracefully via web search fallback

Concept Mapper *Flask · Three.js · OpenRouter API*

January 2026

- Knowledge visualization tool mapping conceptual relationships as an interactive 3D graph; built as a one-week sprint
- LLM-powered concept extraction via OpenRouter API; full-stack Python/Flask backend + vanilla JS + Three.js frontend

EDUCATION

Vellore Institute of Technology

Vellore, India

SKILLS

Languages: Python, C, JavaScript, SQL

AI / ML: Machine Learning, Deep Learning, PyTorch, Retrieval-Augmented Generation (RAG), LangChain, LangGraph, Agentic AI, Model Context Protocol (MCP), Corrective RAG (CRAG), Vector Databases, Fine-Tuning (LoRA+PEFT), Prompt Engineering, Context Engineering

Frameworks & Libraries: FastAPI, Flask, React, Pandas, NumPy, FAISS, ChromaDB, n8n

Tools: Docker, Git, GitHub, Jupyter Notebook, Postgres, MySQL, Three.js, OpenRouter API

ACHIEVEMENTS

- Open Source Contributor — Hugging Face Transformers: fixed incorrect code example in official documentation; PR reviewed and merged by maintainers
- 500K+ LinkedIn impressions through data-driven content strategy